
Toscanini 00 package readme

invr452-fresh-evaporative-vaccine-coolerae7a00faa34a · run run · maturity EXPERIMENT_READY · 3D class ENGINEERING_3D

What this technology is

A recorded invention candidate for the problem: “passive evaporative-cooled vaccine storage vessel for last-mile clinics”. The proposed causal mechanism: • Mechanism: A composite radiative coating embedded with microencapsulated phase-change materials that alter their infrared emissivity based on temperature thresholds

- Intervention: Apply a dual-layer coating where the base layer provides constant radiative cooling while the embedded phase-change layer activates above critical temperature thresholds
- Expected Effect: Dynamic temperature regulati...

The problem addressed

- Problem Id: r452-fresh-evaporative-vaccine-cooler
- Device: passive evaporative-cooled vaccine storage vessel for last-mile clinics
- Failure: vaccine doses spoil in transit because the passive evaporative cooler loses cooling margin within hours when ambient humidity rises, the water-wick channel clogs with dissolved salts, and clinic staff have no indication of how much cooling margin remains before the interior exceeds 8 C
- Failure Mode: cooling_margin_loss
- Constraint: cooler must hold a...

Why the problem matters

reliable operation of passive evaporative-cooled vaccine storage vessel for last-mile clinics

Invention summary

Intervention: • Mechanism: A composite radiative coating embedded with microencapsulated phase-change materials that alter their infrared emissivity based on temperature thresholds

- Intervention: Apply a dual-layer coating where the base layer provides constant radiative cooling while the embedded phase-change l... applied at the recorded intervention site. Status:

EVOLVED_INVENTION_CANDIDATE.

Current maturity

EXPERIMENT_READY —

What is actually demonstrated

The architecture survived the recorded adversarial challenge, adjudication, and evolution stages; the evidence roles are itemized in 04. The parametric engineering model exists with measured geometry and validation gates (ENGINEERING_3D).

What is NOT demonstrated

Nothing in this package was measured in reality; no physical experiment has been run (validation status NOT_POSSIBLE_YET throughout, Art. XXXVIII). EXPERIMENTALLY VERIFIED is not claimed anywhere.

What a buyer receives

This package: the technical essay and engineering dossier, the buyer decision card, the evidence summary with roles, the transfer manifest, the machine-readable engineering layers (traceability, equations, unknown roadmap, validation economics), and the 3D artifacts this invention earned.

Decisive next action

Fund and run the recorded decisive experiment (physical bench/clinical experiment — NOT simulation) — the pre-registered rule either kills the architecture or graduates it to a sourced measurement.

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.